# Engineering Mathematics (AHT-003) - Unit 3

Unit 3 - Calculus III - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

**What this unit covers (syllabus):** Jacobians, approximation of errors, applications of definite integrals (surface areas and volumes of revolution), centre of mass and centre of gravity.

**Key ideas:** The **Jacobian** $J = \partial(u,v)/\partial(x,y)$ is the determinant of partial derivatives; it measures how much a transformation stretches area. Key facts: $J \cdot J' = 1$ (for inverse transformations) and if u, v are functionally dependent then $J = 0$. **Error approximation:** for $z = f(x,y)$, the approximate error $\delta z \approx (\partial f/\partial x)\delta x + (\partial f/\partial y)\delta y$ - used for "percentage error" problems. **Applications of integrals:** area under/between curves, volume of revolution $V = \pi\int y^2 dx$ (about x-axis), and surface area of revolution. **Centre of mass/gravity:** $\bar{x} = \int x\, dm / \int dm$ - for laminas use double integrals, for solids of revolution single integrals with density.

**Exam pointers:** compute a Jacobian (and test functional dependence), one error-approximation numerical (watch percentage-error wording), and a centre-of-gravity problem - the full worked example is in the notes under this unit.

**Extra note - Centre of Gravity by Integration - made simple:**

**Centre of gravity (CG)** = the point where the whole weight feels balanced. Like the middle point of a balance scale.

**Formulas (for a lamina/plate):**

- $\bar{x} = (1/A) \iint x\, dA$ , $\bar{y} = (1/A) \iint y\, dA$
- where $A$ = total area = $\iint dA$

**Steps:** 1) Find the area A with a double integral. 2) Separately find $\iint x\, dA$ and $\iint y\, dA$. 3) Divide each by A - done, you have $(\bar{x}, \bar{y})$.

> *Think of tiny pizza slices. Add up each slice's "x contribution" (x dA), divide by the total area - you get the average x-position. That is the CG's x.*

**Exam tip:** if the shape is symmetric (like a quadrant of a circle or a rectangle), the CG lies on the line of symmetry - half the work done.

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